

FINAL PROJECT · DELOITTE × LUISS 2026

DiscoverAI

Semantic Product Search for Health & Personal Care

A review-aware search system on Amazon Reviews 2023, with quality-driven re-ranking, BART summarization, and domain entity extraction.

Team Mean-Squared-Terrors

THE CHALLENGE

Keyword search doesn't understand what shoppers actually mean

WHAT THE USER TYPES

Natural-language intent
"something to help me sleep at night"

Implicit exclusions
"face cream without parabens"

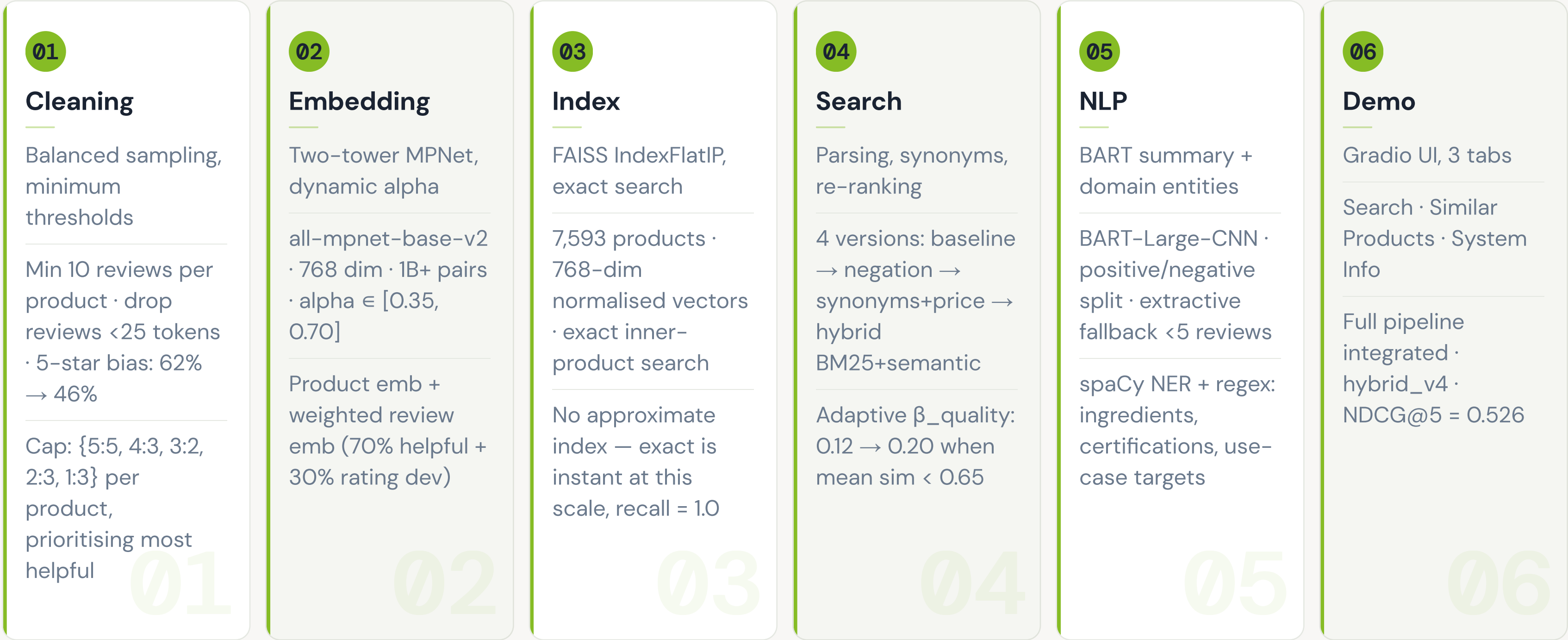
Price intent & trust signals
"professional but affordable shampoo"

WHAT TRADITIONAL SEARCH DOES

- Lexical match: looks for exact words only
- Doesn't understand synonyms or intent
- Cannot tell "with" from "without"
- Ranks by text relevance or raw popularity
- Ignores the customer voice inside reviews

A review-aware system that understands intent and trust

One embedding model, two sources of truth: the product text for what it claims to be, the reviews for how it actually performs, fused into a semantic index with quality-driven re-ranking.



Amazon Reviews 2023: Health & Personal Care

Field coverage, token distribution and reviews per product (balanced set)

7,647

products

318k

cleaned reviews

57k

balanced reviews

96%

brand coverage

CLEANING DECISIONS

- Min 10 reviews per product
Below this threshold the review signal is too weak for a meaningful embedding
- Drop reviews shorter than 25 tokens
"Love it!!", "Fast shipping" carry no semantic content

Product field coverage (%)

Price	26%
Details/Specs	57%
Description	31%
Features	35%
Brand	96%

Product-level overview

Tokens per product (product_text_base)

mean=75

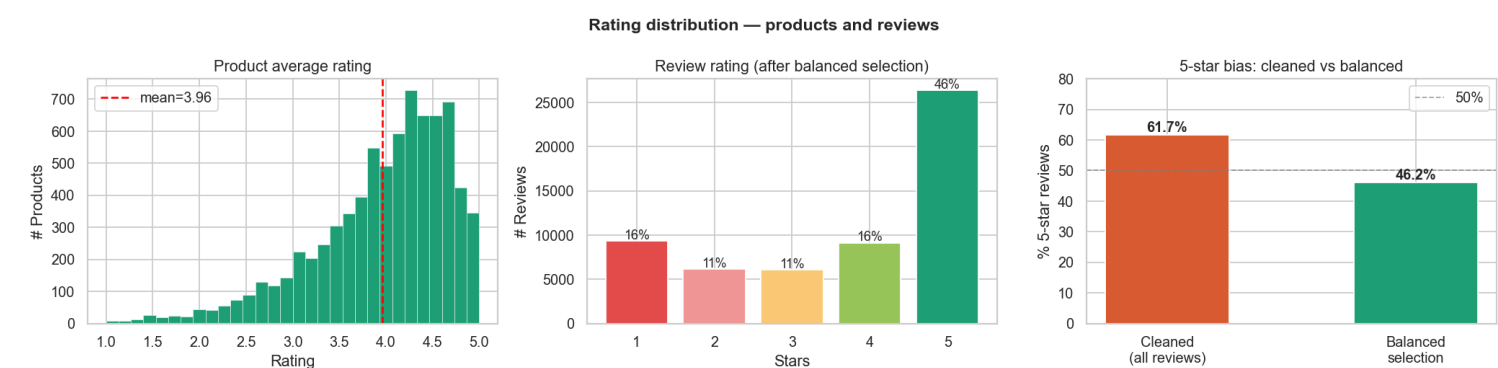
median=38

Reviews per product (balanced)

mean=7.5

REDUCING THE 5-STAR BIAS

Balanced sampling: from 61.7% to 46.2% five-star reviews



WHY THIS MATTERS

Without balancing, the model sees a corpus where 6 out of 10 reviews say "5 stars, love it" — useful signal drowns in praise. The bias correction is per-product, so even niche items keep a critical voice.

HOW THE CAP WORKS

{5:5, 4:3, 3:2, 2:3, 1:3}

Per product, take at most this many reviews per rating, prioritising the most helpful within each band.

Two-tower embedding: product and reviews kept separate

Two sources, two voices

Product text says what the brand sells; reviews say how it actually works. Concatenating lets the noisier source dominate.

Same encoder, two embeddings

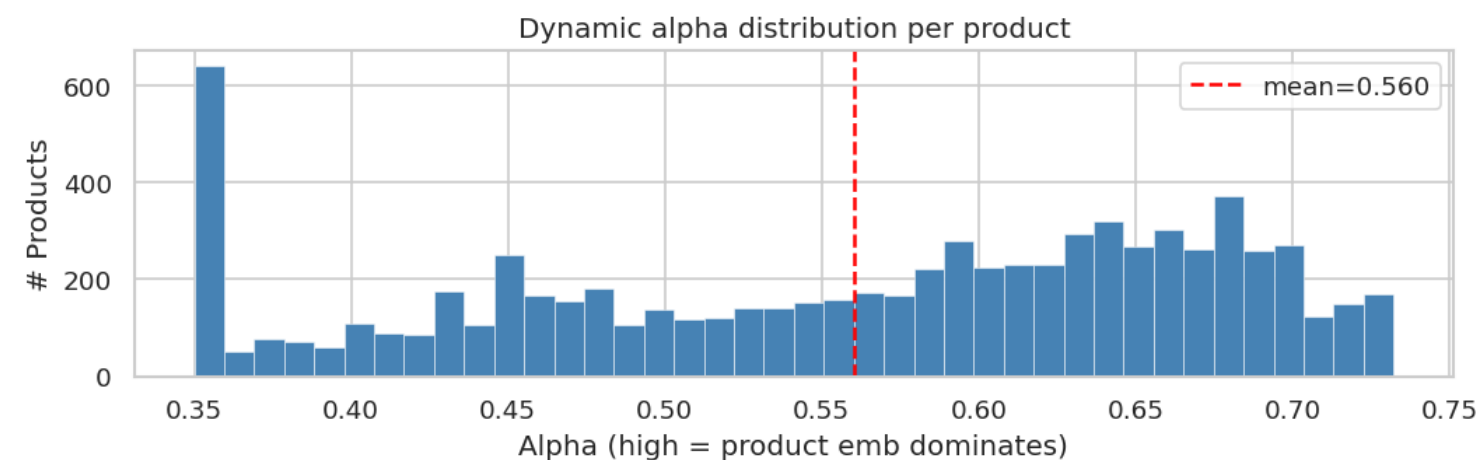
all-mpnet-base-v2 (768 dim, trained on 1B+ pairs): shared semantics, vectors we can weight separately.

Weighted review aggregation

70% helpful_vote score + 30% deviation from neutral rating: most useful reviews surface first.



Dynamic alpha: the product decides the weight, not the engineer



Real distribution of α across the 7,593 products (mean 0.560)

WEAK SIGNAL → HIGH A

Product text dominates

Few reviews, low helpful_vote → product text carries the representation.

$\alpha \approx 0.70$

STRONG SIGNAL → LOW A

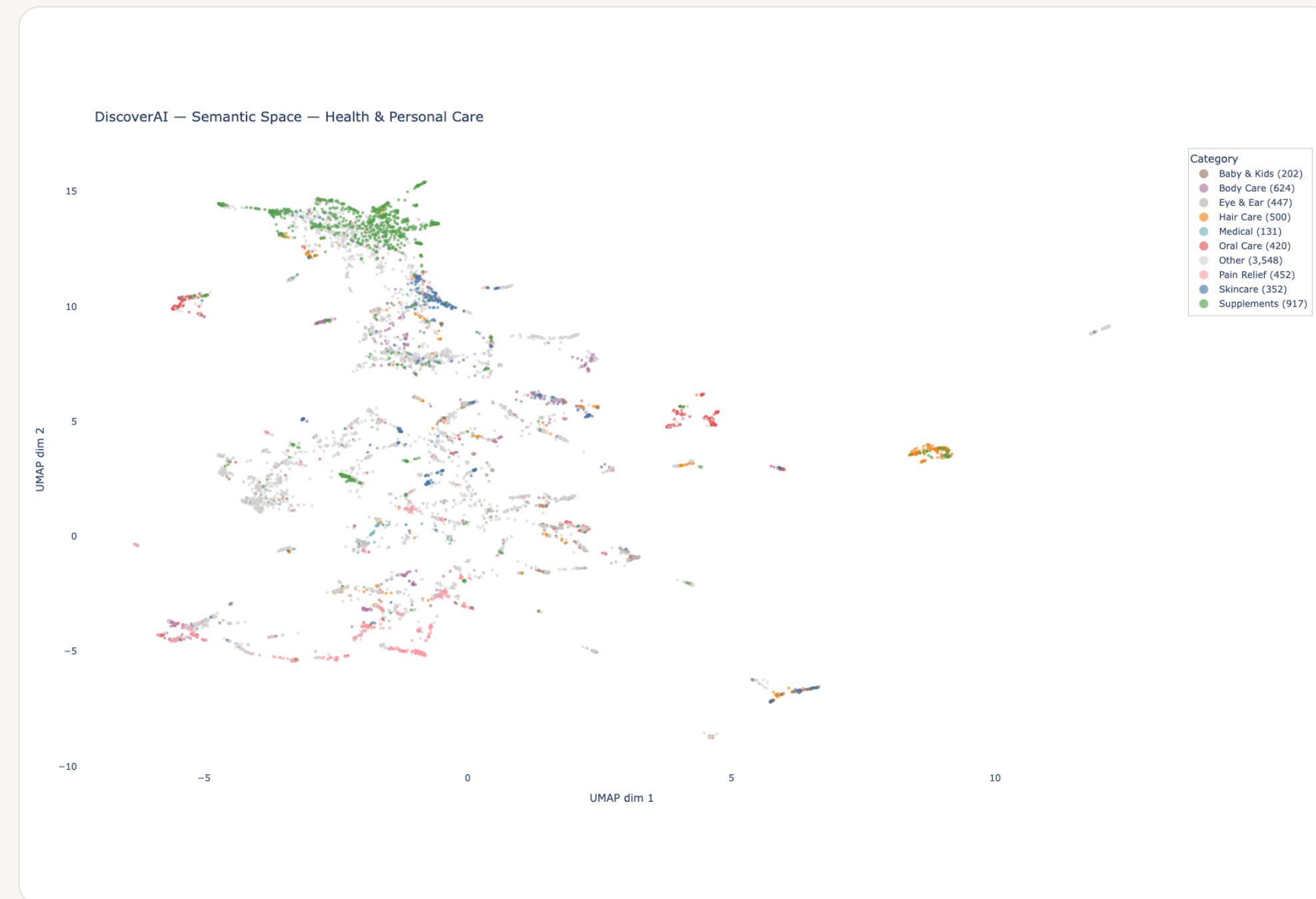
Customer voice carries weight

Many helpful reviews, decisive rating → customer voice carries weight.

$\alpha \approx 0.35$

UMAP projection: semantic clusters emerge by category

7,593 product embeddings · UMAP 2D · coloured by inferred category

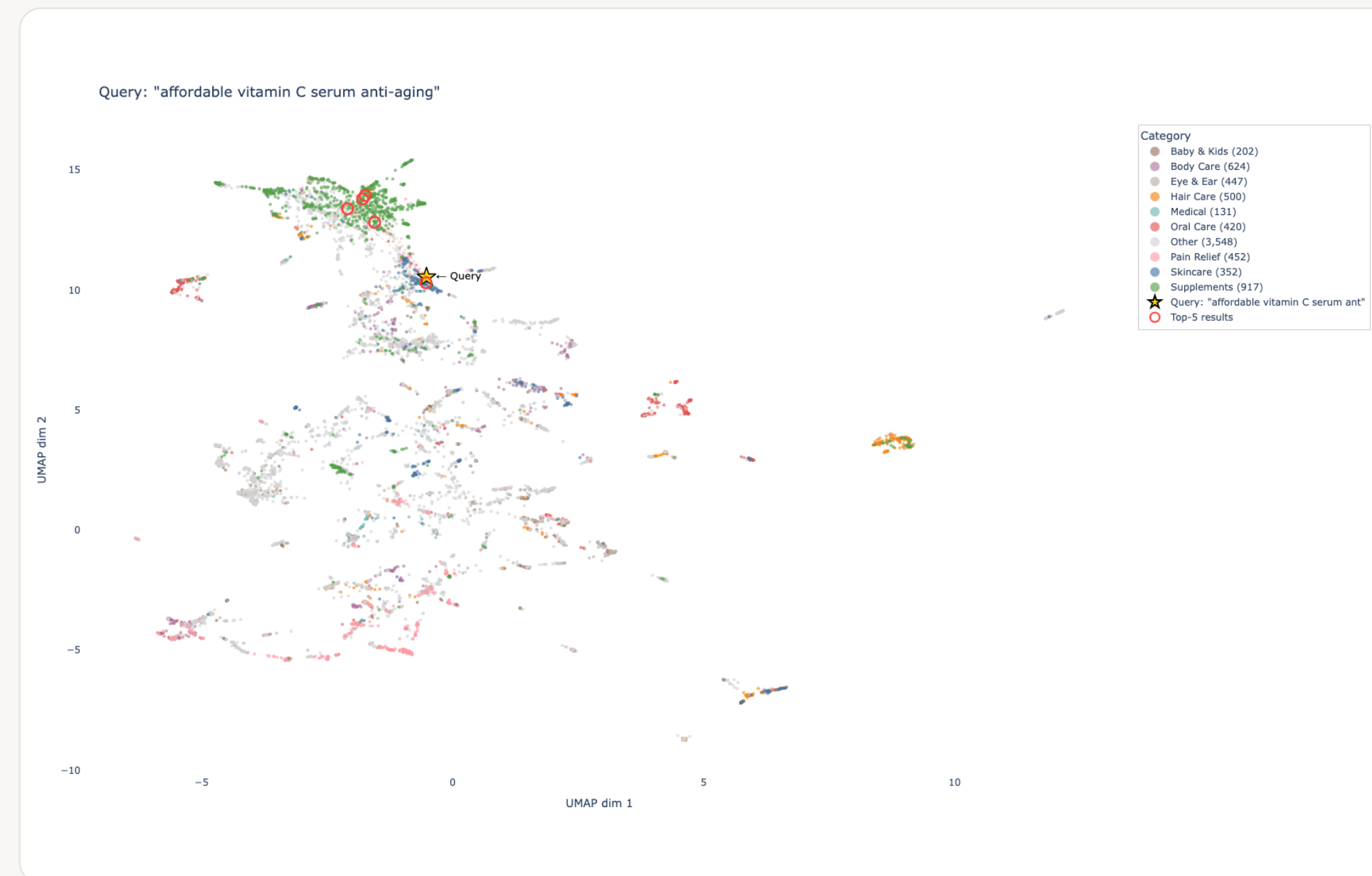


WHAT IT TELLS US

- Supplements form a tight cluster top-left.
- Hair Care separates cleanly from Skin Care.
- Pain Relief sits between supplements and topical products — it bridges both.
- The model learned the catalogue's structure without ever being trained on it: proof of useful semantic geometry.

A query lands exactly where its products live

Query: "affordable vitamin C serum anti-aging" · encoded with same MPNet, projected into same UMAP



WHY THIS WORKS

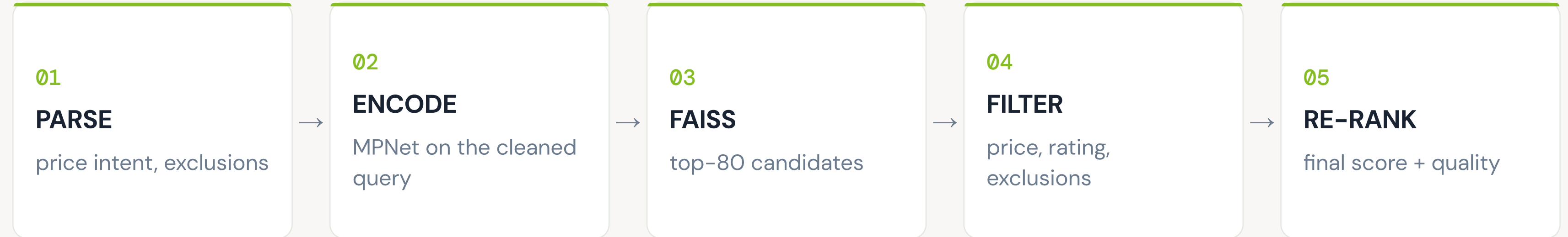
- The query vector falls inside the supplement / skincare neighbourhood, not in a random spot.
- The top-5 results (red circles) are pulled from that same neighbourhood.
- We gain interpretability for free: "why this result?" → because it's spatially close to your intent.

FAISS IndexFlatIP: exact search, no compromises

OPTION	LATENCY @ 7.6K	RECALL	TUNING
IndexFlatIP ✓	negligible	1.0	none
IndexIVF	lower	≈ 0.95	nlist, nprobe to tune
IndexHNSW	lower	≈ 0.97	M, efSearch, slow build

Defense: at 7,593 products exact search is effectively instant. An approximate index trades recall for speed we wouldn't notice. Approximate becomes the right answer once the catalogue grows past **tens of millions**.

From query to ranked results



FINAL SCORE

$\text{score} = \text{similarity} + \beta_{\text{quality}} \cdot \text{quality_score}$

β_{quality} = 0.12 — small enough that a low-quality product cannot overtake a genuinely more relevant one

β_{pop} evaluated but disabled — degraded MRR by ~1.8% in ablation

adaptive β rises to 0.20 when mean similarity < 0.65 (generic query)

QUALITY SCORE

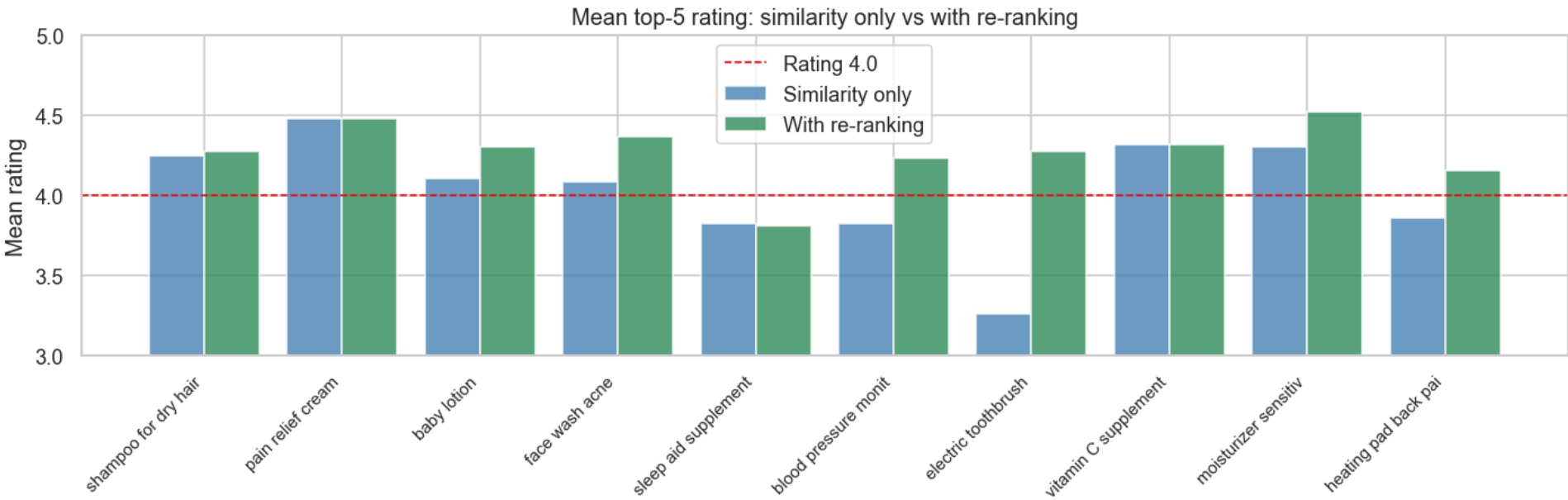
$\text{avg_rating} + \text{frac_positive} + \text{frac_negative} + \log(\text{helpful_votes})$

$0.40 \cdot \text{avg_rating} + \text{fraction positive reviews} + \text{fraction negative reviews} + \log \text{helpful_votes}$, min-max normalised to [0,1]

RE-RANKING PAYS OFF

Mean rating of top-5 results: similarity-only vs with re-ranking

10 representative natural-language queries · $\beta_{\text{quality}}=0.12$



+0.20

stars average lift in top-5

KEY FINDING

Largest gain on noisy / generic queries such as *electric toothbrush* and *sleep aid*.

Iterating through versions: Demo now uses hybrid_v4

V1 • BASELINE

Similarity + fixed re-ranking

The foundation

- FAISS top-50 + score with fixed β
- Works on "catalogue-style" queries
- Fails on negation and natural language

V2 • NEGATION

Filter out what users exclude

Answers observed failure

- Patterns: "without X", "no X", "X-free"
- Extracted terms post-filter the product text
- "cream without parabens" stops returning paraben creams

V3 • PRODUCTION

Synonyms, price, dosages, adaptive β

Full-featured

- Synonym expansion ("SPF" → sunscreen...)
- Price-intent parsing ("affordable" → budget)
- Dosage extraction (1000mg, SPF 50...)
- Adaptive β _quality: 0.20 when mean sim < 0.65

V4 • PRODUCTION (BEST VERSION)

Hybrid v4: BM25 + Semantic + RRF

Powers the demo

- Demo powered by hybrid_v4
- BM25 + Semantic + Reciprocal Rank Fusion
- Evaluation — NDCG@5 = 0.526

SUMMARIZATION & EXTRACTION

From thousands of reviews to a readable card in seconds

SUMMARIZATION

BART-Large-CNN

- Split into positive ($\geq 4\star$) and negative ($\leq 2\star$)
- Filters: ≥ 12 words, no "I...", < 300 chars, deduped
- BART produces summary + pros + cons
- Extractive fallback if < 5 reviews — avoids hallucinations on thin evidence

ENTITY RECOGNITION

Domain regex + spaCy NER

ACTIVE INGREDIENTS

retinol niacinamide hyaluronic acid SPF omega-3

CERTIFICATIONS

organic vegan cruelty-free paraben-free

USE-CASE TARGETS

skin type hair type age group body area

Gradio interface, 3 tabs, the entire pipeline integrated

01

Search

Free-text query with optional filters for price bucket and minimum rating.

- product image, rating, price
- BART summary (pros & cons)
- extracted active ingredients & certifications
- powered by hybrid_v4 (BM25 + semantic + RRF)
- negation, synonym expansion, price-intent parsing

02

Similar Products

Content-based recommendation: from a product name, returns the 5 most similar on the combined embedding.

- no text encoding at runtime
- operates directly on combined_emb matrix
- instant lookup via FAISS inner-product
- shows image, rating, price, BART summary

03

System Info

System transparency for evaluators: index size, model in use, feature coverage.

- index stats (7,593 products · 768-dim)
- model: all-mpnet-base-v2 + BART-Large-CNN
- feature coverage (summaries, entities)
- evaluation: NDCG@5 = 0.526 (hybrid_v4)

The architectural decisions we would make again

Two-tower over concatenation

Mixing product and review text in one 512-token sequence lets the noisier source dominate. Two towers let us weight contributions explicitly.

Dynamic α over fixed α

An average weight punishes products with credible review evidence and inflates those with thin reviews. Per-product α respects both ends.

Exact FAISS over approximate

At 7,593 products, exact is effectively instant and recall = 1. We adopt approximate when the catalogue forces it, not before.

Iteration v1→v2→v3→v4 over big bang

Each version answers an observed failure (negation, synonyms, hybrid fusion). Clear responsibilities and easy regression testing.

Balanced sampling + 10-review threshold

Cuts the 5-star bias from 62% to 46% and excludes products whose review signal is too thin to generate a reliable embedding.

Extractive fallback for BART

On few reviews, abstractive summarization hallucinates. Extractive is less elegant but honest about the evidence.

MEAN-SQUARED-TERRORS · DISCOVERAI · DELOITTE × LUISS 2026

Thank You

Questions